Emin Erkan YILDIRIM

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Education

Dokuz Eylul University – MS in Electrical and Electronics Engineering — February 2025 – Present
Dokuz Eylul University – BS in Electrical and Electronics Engineering (GPA: 3.0/4.0) — September 2019 – June 2024

Experience

Laboratory Specialist R&D, Vestel Beyaz Eşya – Manisa, Türkiye — October 2024 – Present

- Verify tumble-dryer electronic control boards and investigate faults using error logs, oscilloscopes, and multimeters.
- Track and coordinate prototype builds and board bring-up, and support board-level test and verification.
- Work with software and firmware teams on hardware–firmware integration.
- Write test procedures based on applicable IEC 60335 requirements, prepare engineering requirements specifications and hardware-development documentation, and report test results.
- Benchmark competitor products and coordinate with suppliers to evaluate alternative components for technical suitability, cost, quality, and availability.

Embedded Software Engineering Intern, Baykar Technologies – İstanbul, Türkiye — July 2023 – August 2023

- Studied Data Distribution Service (DDS), compared it with alternative communication protocols, and presented the technical findings.
- Developed a basic Python application to simulate DDS communication.

Research and Development Intern, Güleryüz Karoser Otomotiv A.Ş. – Bursa, Türkiye — July 2022 – August 2022

- Reviewed CAN bus routing in an electric-bus electrical-system model and determined required cable lengths.
- Standardized signal names in the model to support production-line installation.

Selected Engineering Projects

M.Sc. Thesis: RISC-V Affine-Arithmetic Processing Unit on FPGA — In progress

- Designing an FPGA hardware unit, driven by a RISC-V core through custom instructions, to speed up uncertainty and tolerance analysis of engineering systems.

Smart Sportswear: BLE ECG & Motion Sensor (TÜBİTAK 1001) — 2023

- Designed and revised a compact KiCad PCB for a garment-integrated ECG and motion-sensing system using conductive-yarn electrodes.
- Identified an ECG interface voltage-level mismatch during board bring-up, reworked the prototype, and incorporated the correction into a PCB revision.
- Developed RSL10 application firmware in C for ECG sampling, MPU-6050 accelerometer acquisition over I2C, and data transmission using BLE GATT notifications.
- Developed a C tool for decoding BLE logs into CSV files and MATLAB scripts for ECG peak detection and heart-rate estimation.

Solar-Car Telemetry, CAN Communication & Data Logging (Solar Team Solaris) — 2022

- Developed CAN communication software in C on an STM32 microcontroller and built circuitry for communication testing and validation.
- Analyzed LoRa telemetry frames using HTerm and developed a C# application to receive packets through a USB-connected LoRa module and parse their contents into categorized data.
- Implemented packet-checksum validation, live visualization with LiveCharts, and CSV logging; operated it during an actual race for over one hour without observed crashes or freezes.

Skills

Programming Languages: C, C#, Python, SystemVerilog, MATLAB
Tools: KiCad, LTspice, QUCS, Proteus, Webots
Communication Protocols: TCP/IP, UDP, SPI, UART, CAN, I2C, BLE, MIL-STD-1553
Languages: English (Professional working proficiency), Turkish (Native)